

Expanding Brackets

Worked Solutions with Area-Model Diagrams

Wisest Maths — Year 1 A-Level, Algebra (ref **a1**)

This document contains all 45 *Expanding Brackets* questions, each with a fully worked solution and a TikZ *area-model* (box-method) diagram. In an area model the terms of one bracket label the rows, the terms of the other label the columns, and each interior cell holds the product of its row and column headings; summing the cells (and collecting like terms) gives the expansion.

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Expanding Brackets 01 (a1-001)

Difficulty: Foundation / Marks: 2

Expand the brackets $(x + 4)(x - 1)$.

Worked solution.

Expand and simplify

Expanding the brackets:

$$\begin{aligned}
 (x + 4)(x - 1) &= x(x - 1) + 4(x - 1) \\
 &= x^2 - x + 4x - 4 \\
 &= x^2 + 3x - 4
 \end{aligned}$$

Multiply each term in the first bracket by each term in the second, then collect like terms.

Final answer: $x^2 + 3x - 4$

Common mistake. A common mistake is to forget to multiply the 4 by the -1 , which gives the incorrect answer $x^2 + 3x + 4$.

Area-model diagram.

$\times$	x	-1
x	x^2	$-x$
4	$4x$	-4

Expanding Brackets 02 (a1-002)

Difficulty: Foundation / Marks: 2

Expand the brackets $(2x + 1)(x - 3)$.

Worked solution.

Expand and simplify

Expanding the brackets:

$$\begin{aligned}(2x + 1)(x - 3) &= 2x(x - 3) + 1(x - 3) \\ &= 2x^2 - 6x + x - 3 \\ &= 2x^2 - 5x - 3\end{aligned}$$

Multiply each term in the first bracket by each term in the second, then collect like terms.

Final answer: $2x^2 - 5x - 3$

Common mistake. A common mistake is to forget to multiply the -3 by the 1 , which gives the incorrect answer $2x^2 - 6x - 3$.

Area-model diagram.

$\times$	x	-3
$2x$	$2x^2$	$-6x$
1	x	-3

Expanding Brackets 03 (a1-003)

Difficulty: Foundation / Marks: 2

Expand the brackets $(x + 1)(x - 2)(x + 3)$.

Worked solution.

Expand the last two brackets first, then multiply by the remaining bracket

Expanding the last two brackets:

$$\begin{aligned}(x + 1)((x - 2)(x + 3)) &= (x + 1)(x^2 + 3x - 2x - 6) \\ &= (x + 1)(x^2 + x - 6)\end{aligned}$$

Now expand:

$$\begin{aligned}(x + 1)(x^2 + x - 6) &= x(x^2 + x - 6) + 1(x^2 + x - 6) \\ &= x^3 + x^2 - 6x + x^2 + x - 6 \\ &= x^3 + 2x^2 - 5x - 6\end{aligned}$$

First expand two of the three brackets, then multiply the result by the third bracket.

Final answer: $x^3 + 2x^2 - 5x - 6$

Area-model diagram.

$\times$	x^2	x	-6
x	x^3	x^2	$-6x$
1	x^2	x	-6

After expanding the last two brackets to $x^2 + x - 6$, multiply by $(x + 1)$.

Expanding Brackets 04 (a1-004)

Difficulty: Foundation / Marks: 2

Expand $xy^3(x + y)(x^2 + 2xy - 3y^2)$.

Worked solution.

Expand the inner brackets first, then multiply by the outer factor

Expand the inner brackets:

$$\begin{aligned}
 xy^3(x(x^2 + 2xy - 3y^2) + y(x^2 + 2xy - 3y^2)) &= xy^3(x^3 + 2x^2y - 3xy^2 + x^2y + 2xy^2 - 3y^3) \\
 &= xy^3(x^3 + 3x^2y - xy^2 - 3y^3)
 \end{aligned}$$

Now distribute xy^3 :

$$= x^4y^3 + 3x^3y^4 - x^2y^5 - 3xy^6$$

Expand the two brackets in parentheses first, collect like terms, then multiply every term by the outer factor.

Final answer: $x^4y^3 + 3x^3y^4 - x^2y^5 - 3xy^6$

Area-model diagram.

$\times$	x^2	$2xy$	$-3y^2$
x	x^3	$2x^2y$	$-3xy^2$
y	x^2y	$2xy^2$	$-3y^3$

Grid for $(x + y)(x^2 + 2xy - 3y^2)$; every term is then multiplied by the outer factor xy^3 .

Expanding Brackets 05 (a1-005)

Difficulty: Foundation / Marks: 4

Expand $\left(x + xy + \frac{x^2}{y}\right)\left(x^2 - \frac{y}{x^2} + y^2\right)$.

Worked solution.

Expand the brackets

Expand the brackets:

$$\begin{aligned} & \left(x + xy + \frac{x^2}{y}\right)\left(x^2 - \frac{y}{x^2} + y^2\right) \\ &= x\left(x^2 - \frac{y}{x^2} + y^2\right) + xy\left(x^2 - \frac{y}{x^2} + y^2\right) + \frac{x^2}{y}\left(x^2 - \frac{y}{x^2} + y^2\right) \\ &= x^3 - \frac{y}{x} + xy^2 + x^3y - \frac{y^2}{x} + xy^3 + \frac{x^4}{y} - 1 + x^2y \\ &= x^3 + x^3y + x^2y + xy^2 + xy^3 + \frac{x^4}{y} - \frac{y}{x} - \frac{y^2}{x} - 1 \end{aligned}$$

Multiply each term in the first bracket by each term in the second, then collect like terms.

Final answer: $x^3 + x^3y + x^2y + xy^2 + xy^3 + \frac{x^4}{y} - \frac{y}{x} - \frac{y^2}{x} - 1$

Area-model diagram.

$\times$	x^2	$-\frac{y}{x^2}$	y^2
x	x^3	$-\frac{y}{x}$	xy^2
xy	x^3y	$-\frac{y^2}{x}$	xy^3
$\frac{x^2}{y}$	$\frac{x^4}{y}$	-1	x^2y

Expanding Brackets 06 (a1-006)

Difficulty: Foundation / Marks: 4

Expand $4x - \left(3(x - 2) - 4(2x + 3(x - 3))\right)$.

Worked solution.

Expand the brackets

Expand the brackets:

$$\begin{aligned} & 4x - (3(x - 2) - 4(2x + 3(x - 3))) \\ &= 4x - (3x - 6 - 4(2x + 3x - 9)) \\ &= 4x - (3x - 6 - 4(5x - 9)) \\ &= 4x - (3x - 6 - 20x + 36) \\ &= 4x - (-17x + 30) \\ &= 4x + 17x - 30 \\ &= 21x - 30 \end{aligned}$$

Start by expanding the innermost brackets, then work outwards, simplifying at each step.

Final answer: $21x - 30$

Area-model diagram.

$\times$	$5x$	-9
-4	$-20x$	36

Key step: distributing -4 over the simplified inner bracket $(5x - 9)$.

Expanding Brackets 07 (a1-007)

Difficulty: Foundation / Marks: 4

Expand $(a + b - c)(a - b + c) + (b + c - a)(b - c + a)$.

Worked solution.

Expand the brackets

$$\begin{aligned} & (a + b - c)(a - b + c) + (b + c - a)(b - c + a) \\ &= (a + (b - c))(a - (b - c)) + (b + (c - a))(b - (c - a)) \\ &= a^2 - (b - c)^2 + b^2 - (c - a)^2 \\ &= a^2 - (b^2 - 2bc + c^2) + b^2 - (c^2 - 2ac + a^2) \\ &= a^2 - b^2 + 2bc - c^2 + b^2 - c^2 + 2ac - a^2 \\ &= 2bc - 2c^2 + 2ac \\ &= 2c(a + b - c) \end{aligned}$$

Recognise that $(a+b-c)(a-b+c) = a^2 - (b-c)^2$ using the difference of two squares. Similarly for the second pair. Then expand and collect like terms.

Final answer: $2c(a + b - c)$

Area-model diagram.

$\times$	a	$-b$	c
a	a^2	$-ab$	ac
b	ab	$-b^2$	bc
$-c$	$-ac$	bc	$-c^2$

Grid for the first product $(a + b - c)(a - b + c)$; the second product is handled the same way.

Expanding Brackets 08 (a1-008)

Difficulty: Foundation / Marks: 4

Expand $(2x - 3y^2)^3$.

Worked solution.

Expand step by step, showing each term multiplying out

$$\begin{aligned}
 (2x - 3y^2)^3 &= (2x - 3y^2) \left[(2x - 3y^2)(2x - 3y^2) \right] \\
 &= (2x - 3y^2) \left[2x(2x - 3y^2) - 3y^2(2x - 3y^2) \right] \\
 &= (2x - 3y^2) \left[4x^2 - 6xy^2 - 6xy^2 + 9y^4 \right] \\
 &= (2x - 3y^2)(4x^2 - 12xy^2 + 9y^4) \\
 &= 2x(4x^2 - 12xy^2 + 9y^4) - 3y^2(4x^2 - 12xy^2 + 9y^4) \\
 &= 8x^3 - 24x^2y^2 + 18xy^4 - 12x^2y^2 + 36xy^4 - 27y^6 \\
 &= 8x^3 - 36x^2y^2 + 54xy^4 - 27y^6
 \end{aligned}$$

First expand the inner two brackets showing $a(c+d)+b(c+d)$, collect like terms, then multiply the result by the remaining bracket using the same method.

Final answer: $8x^3 - 36x^2y^2 + 54xy^4 - 27y^6$

Area-model diagram.

$\times$	$2x$	$-3y^2$
$2x$	$4x^2$	$-6xy^2$
$-3y^2$	$-6xy^2$	$9y^4$

Grid for $(2x - 3y^2)^2$; the result $(4x^2 - 12xy^2 + 9y^4)$ is then multiplied by $(2x - 3y^2)$ again.

Expanding Brackets 09 (a1-009)

Difficulty: Foundation / Marks: 4

Expand $(m + n)(m^3 - mn^2 + n^3)$.

Worked solution.

Expand the brackets

Expand the brackets:

$$\begin{aligned}(m + n)(m^3 - mn^2 + n^3) &= m(m^3 - mn^2 + n^3) + n(m^3 - mn^2 + n^3) \\ &= m^4 - m^2n^2 + mn^3 + m^3n - mn^3 + n^4 \\ &= m^4 + m^3n - m^2n^2 + n^4\end{aligned}$$

Expand the first term in the first bracket with the last bracket, and then the second term in the first bracket with the last bracket.

Final answer: $m^4 + m^3n - m^2n^2 + n^4$

Area-model diagram.

$\times$	m^3	$-mn^2$	n^3
m	m^4	$-m^2n^2$	mn^3
n	m^3n	$-mn^3$	n^4

Expanding Brackets 10 (a1-010)

Difficulty: Foundation / Marks: 4

Expand $pq(q^2 - 2p + 3q) + rs(r^2 - 2s + 3r)$.

Worked solution.

Expand the brackets

Expand the brackets:

$$pq(q^2 - 2p + 3q) + rs(r^2 - 2s + 3r) = pq^3 - 2p^2q + 3pq^2 + r^3s - 2rs^2 + 3r^2s$$

Expand each bracket separately, then collect like terms.

Final answer: $pq^3 - 2p^2q + 3pq^2 + r^3s - 2rs^2 + 3r^2s$

Area-model diagram.

$\times$	q^2	$-2p$	$3q$
pq	pq^3	$-2p^2q$	$3pq^2$

Grid for $pq(q^2 - 2p + 3q)$; the term $rs(r^2 - 2s + 3r)$ expands identically.

Expanding Brackets 11 (a1-011)

Difficulty: Foundation / Marks: 4

Expand $(x + y + z)(x - y - z)$.

Worked solution.

Expand the brackets

Expand the brackets:

$$\begin{aligned}(x + y + z)(x - y - z) &= x(x - y - z) + y(x - y - z) + z(x - y - z) \\ &= x^2 - xy - xz + xy - y^2 - yz + xz - yz - z^2 \\ &= x^2 - y^2 - z^2 - 2yz\end{aligned}$$

Expand each bracket separately, then collect like terms.

Final answer: $x^2 - y^2 - z^2 - 2yz$

Area-model diagram.

$\times$	x	$-y$	$-z$
x	x^2	$-xy$	$-xz$
y	xy	$-y^2$	$-yz$
z	xz	$-yz$	$-z^2$

Expanding Brackets 12 (a1-012)

Difficulty: Foundation / Marks: 4

Expand $(x + 1)(2x - 1)^2 + (x - 1)(3x - 2)^2$.

Worked solution.

Expand the brackets

$$\begin{aligned}&(x + 1)(2x - 1)^2 + (x - 1)(3x - 2)^2 \\ &= (x + 1)(2x(2x - 1) - 1(2x - 1)) + (x - 1)(3x(3x - 2) - 2(3x - 2)) \\ &= (x + 1)(4x^2 - 4x + 1) + (x - 1)(9x^2 - 12x + 4) \\ &= x(4x^2 - 4x + 1) + 1(4x^2 - 4x + 1) + x(9x^2 - 12x + 4) - 1(9x^2 - 12x + 4) \\ &= (4x^3 - 4x^2 + x + 4x^2 - 4x + 1) + (9x^3 - 12x^2 + 4x - 9x^2 + 12x - 4) \\ &= (4x^3 - 3x + 1) + (9x^3 - 21x^2 + 16x - 4) \\ &= 13x^3 - 21x^2 + 13x - 3\end{aligned}$$

Expand each squared bracket using $a(c+d)+b(c+d)$, then multiply out and collect like terms.

Final answer: $13x^3 - 21x^2 + 13x - 3$

Area-model diagram.

$\times$	$2x$	-1
$2x$	$4x^2$	$-2x$
-1	$-2x$	1

Grid for $(2x - 1)^2$; expand $(3x - 2)^2$ the same way, then multiply each by its linear factor.

Expanding Brackets 13 (a1-013)

Difficulty: Foundation / Marks: 4

Expand $(x + 2)(3x + 1)^2 - (x - 1)(2x + 3)^2$.

Worked solution.

Expand each squared bracket, then multiply out

$$\begin{aligned}
 & (x + 2)(3x + 1)^2 - (x - 1)(2x + 3)^2 \\
 &= (x + 2)(3x(3x + 1) + 1(3x + 1)) - (x - 1)(2x(2x + 3) + 3(2x + 3)) \\
 &= (x + 2)(9x^2 + 6x + 1) - (x - 1)(4x^2 + 12x + 9) \\
 &= x(9x^2 + 6x + 1) + 2(9x^2 + 6x + 1) - [x(4x^2 + 12x + 9) - 1(4x^2 + 12x + 9)] \\
 &= (9x^3 + 6x^2 + x + 18x^2 + 12x + 2) - (4x^3 + 12x^2 + 9x - 4x^2 - 12x - 9) \\
 &= (9x^3 + 24x^2 + 13x + 2) - (4x^3 + 8x^2 - 3x - 9) \\
 &= 9x^3 + 24x^2 + 13x + 2 - 4x^3 - 8x^2 + 3x + 9 \\
 &= 5x^3 + 16x^2 + 16x + 11
 \end{aligned}$$

Expand each squared bracket using $a(c+d)+b(c+d)$, multiply out both products, then subtract and collect like terms.

Final answer: $5x^3 + 16x^2 + 16x + 11$

Area-model diagram.

$\times$	$3x$	1
$3x$	$9x^2$	$3x$
1	$3x$	1

Grid for $(3x + 1)^2$; treat $(2x + 3)^2$ the same way.

Expanding Brackets 14 (a1-014)

Difficulty: Foundation / Marks: 4

Expand $(3a + 2b)^2 - (3a - 2b)^2$.

Worked solution.

Expand the squares and simplify

Expand the brackets:

$$\begin{aligned}(3a + 2b)^2 - (3a - 2b)^2 &= (3a + 2b)(3a + 2b) - (3a - 2b)(3a - 2b) \\&= 3a(3a + 2b) + 2b(3a + 2b) - 3a(3a - 2b) - 2b(3a - 2b) \\&= 9a^2 + 6ab + 6ab + 4b^2 - (9a^2 - 6ab - 6ab + 4b^2) \\&= 9a^2 + 12ab + 4b^2 - (9a^2 - 12ab + 4b^2) \\&= 9a^2 + 12ab + 4b^2 - 9a^2 + 12ab - 4b^2 \\&= 24ab\end{aligned}$$

Expand both squared binomials separately, then subtract the second expansion from the first, taking care with the signs.

Final answer: $24ab$

Area-model diagram.

$\times$	$3a$	$2b$
$3a$	$9a^2$	$6ab$
$2b$	$6ab$	$4b^2$

Grid for $(3a + 2b)^2$; subtract the analogous grid for $(3a - 2b)^2$.

Expanding Brackets 15 (a1-015)

Difficulty: Foundation / Marks: 4

Expand $(x^2 + 2)(x^2 - 5x + 3)$.

Worked solution.

Expand the brackets

Expand the brackets:

$$\begin{aligned}
 (x^2 + 2)(x^2 - 5x + 3) &= x^2(x^2 - 5x + 3) + 2(x^2 - 5x + 3) \\
 &= x^2(x^2) + x^2(-5x) + x^2(3) + 2(x^2) + 2(-5x) + 2(3) \\
 &= x^4 - 5x^3 + 3x^2 + 2x^2 - 10x + 6 \\
 &= x^4 - 5x^3 + 5x^2 - 10x + 6
 \end{aligned}$$

Distribute each term in the first bracket across the second bracket, then combine the like terms.

Final answer: $x^4 - 5x^3 + 5x^2 - 10x + 6$

Area-model diagram.

$\times$	x^2	$-5x$	3
x^2	x^4	$-5x^3$	$3x^2$
2	$2x^2$	$-10x$	6

Expanding Brackets 16 (a1-016)

Difficulty: Foundation / Marks: 4

Expand $(a - b)(a + b)(a^2 + b^2)$.

Worked solution.

Expand the first two brackets

Step 1: Difference of two squares

$$\begin{aligned}
 (a - b)(a + b)(a^2 + b^2) &= (a(a + b) - b(a + b))(a^2 + b^2) \\
 &= (a^2 + ab - ba - b^2)(a^2 + b^2) \\
 &= (a^2 - b^2)(a^2 + b^2) \\
 &= (a^2(a^2 + b^2) - b^2(a^2 + b^2)) \\
 &= a^4 + a^2b^2 - b^2a^2 - b^4 \\
 &= a^4 - b^4
 \end{aligned}$$

Recognize that the first two brackets form a difference of two squares.

Final answer: $a^4 - b^4$

Area-model diagram.

$\times$	a^2	b^2
a^2	a^4	a^2b^2
$-b^2$	$-a^2b^2$	$-b^4$

After $(a - b)(a + b) = a^2 - b^2$, this grid multiplies it by $(a^2 + b^2)$.

Expanding Brackets 17 (a1-017)

Difficulty: Foundation / Marks: 4

Expand $2x(x - 3)^2 - x^2(2x + 1)$.

Worked solution.

Expand the squares and distribute

Expand the brackets:

$$\begin{aligned}
 2x(x - 3)^2 - x^2(2x + 1) &= 2x(x - 3)(x - 3) - (2x^3 + x^2) \\
 &= 2x(x(x - 3) - 3(x - 3)) - (2x^3 + x^2) \\
 &= 2x(x^2 - 3x - 3x + 9) - (2x^3 + x^2) \\
 &= 2x(x^2 - 6x + 9) - (2x^3 + x^2) \\
 &= 2x^3 - 12x^2 + 18x - 2x^3 - x^2 \\
 &= -13x^2 + 18x
 \end{aligned}$$

First square the binomial, then multiply by 2x. Expand the second part and carefully subtract to collect like terms.

Final answer: $-13x^2 + 18x$

Area-model diagram.

$\times$	x	-3
x	x^2	$-3x$
-3	$-3x$	9

Grid for $(x - 3)^2$; the result is then multiplied by $2x$.

Expanding Brackets 18 (a1-018)

Difficulty: Foundation / Marks: 4

Expand $(x + 2)(x - 3)(x + 4)$.

Worked solution.

Expand the first two brackets

Expand:

$$\begin{aligned}(x + 2)((x - 3)(x + 4)) &= (x + 2)(x(x + 4) - 3(x + 4)) \\&= (x + 2)(x^2 + 4x - 3x - 12) \\&= (x + 2)(x^2 + x - 12) \\&= x(x^2 + x - 12) + 2(x^2 + x - 12) \\&= x^3 + x^2 - 12x + 2x^2 + 2x - 24 \\&= x^3 + 3x^2 - 10x - 24\end{aligned}$$

Multiply the first two terms and simplify.

Final answer: $x^3 + 3x^2 - 10x - 24$

Area-model diagram.

$\times$	x^2	x	-12
x	x^3	x^2	$-12x$
2	$2x^2$	$2x$	-24

After $(x - 3)(x + 4) = x^2 + x - 12$, this grid multiplies it by $(x + 2)$.

Expanding Brackets 19 (a1-019)

Difficulty: Foundation / Marks: 3

A rectangle has a length of $2x + 5$ and a width of $x - 3$. Write an expanded expression for the area of the rectangle.

Worked solution.

Set up and expand

$$\begin{aligned}\text{Area} &= (2x + 5)(x - 3) \\&= 2x(x - 3) + 5(x - 3) \\&= 2x^2 - 6x + 5x - 15 \\&= 2x^2 - x - 15\end{aligned}$$

The area of a rectangle is length times width. Expand the brackets and collect like terms.

Final answer: $2x^2 - x - 15$

Area-model diagram.

$\times$	x	-3
$2x$	$2x^2$	$-6x$
5	$5x$	-15

This is literally the rectangle: length $2x + 5$ down the side, width $x - 3$ along the top.

Expanding Brackets 20 (a1-020)

Difficulty: Foundation / Marks: 3

Expand and simplify $\frac{1}{4}(12x - 8) - \frac{1}{5}(10x + 5)$.

Worked solution.

Expand the brackets using the fractional coefficients

Expand:

$$\begin{aligned}\frac{1}{4}(12x - 8) - \frac{1}{5}(10x + 5) &= 3x - 2 - 2x - 1 \\ &= x - 3\end{aligned}$$

Divide the first bracket by 4 and the second bracket by 5, being careful to distribute the negative sign.

Final answer: $x - 3$

Area-model diagram.

$\times$	$12x$	-8
$\frac{1}{4}$	$3x$	-2

Grid for $\frac{1}{4}(12x - 8)$; subtract the analogous grid for $\frac{1}{5}(10x + 5) = 2x + 1$.

Expanding Brackets 21 (a1-021)

Difficulty: Foundation / Marks: 4

Expand and simplify $2x(x + 3) - (x + 1)(x + 2)$.

Worked solution.

Expand the single and double brackets

$$\begin{aligned}2x(x + 3) - (x + 1)(x + 2) &= 2x^2 + 6x - x(x + 2) - 1(x + 2) \\ &= 2x^2 + 6x - x^2 - 2x - x - 2 \\ &= x^2 + 3x - 2\end{aligned}$$

Expand the first part by distributing $2x$, and expand the brackets of the second part.

Final answer: $x^2 + 3x - 2$

Area-model diagram.

$\times$	x	2
x	x^2	$2x$
1	x	2

Grid for $(x+1)(x+2)$; this product is subtracted from $2x(x+3) = 2x^2 + 6x$.

Expanding Brackets 22 (a1-022)

Difficulty: Foundation / Marks: 4

Expand and simplify $(a+b)^2 - (a-b)^2$.

Worked solution.

Expand both squared brackets

$$\begin{aligned}
 (a+b)^2 - (a-b)^2 &= (a+b)(a+b) - (a-b)(a-b) \\
 &= a(a+b) + b(a+b) - [a(a-b) - b(a-b)] \\
 &= a^2 + ab + ab + b^2 - (a^2 - ab - ab + b^2) \\
 &= a^2 + 2ab + b^2 - a^2 + 2ab - b^2 \\
 &= 4ab
 \end{aligned}$$

Expand each squared bracket using $a(c+d)+b(c+d)$, then subtract and collect like terms.

Final answer: $4ab$

Area-model diagram.

$\times$	a	b
a	a^2	ab
b	ab	b^2

Grid for $(a+b)^2$; subtract the analogous grid for $(a-b)^2$.

Expanding Brackets 23 (a1-023)

Difficulty: Foundation / Marks: 3

Expand $x[3(x + 2) - 4]$.

Worked solution.

Simplify the expression inside the square brackets first

$$\begin{aligned}x(3(x + 2) - 4) &= x(3x + 6 - 4) \\&= x(3x + 2) \\&= 3x^2 + 2x\end{aligned}$$

Start from the inner-most brackets and work outward.

Final answer: $3x^2 + 2x$

Area-model diagram.

$\times$	$3x$	2
x	$3x^2$	$2x$

After simplifying the inside to $(3x + 2)$, this grid multiplies by x .

Expanding Brackets 24 (a1-024)

Difficulty: Foundation / Marks: 4

Expand and simplify $(x + \frac{1}{x^2})(y - \frac{1}{z^2})$.

Worked solution.

Expand the brackets

$$\begin{aligned}\left(x + \frac{1}{x^2}\right)\left(y - \frac{1}{z^2}\right) &= x\left(y - \frac{1}{z^2}\right) + \frac{1}{x^2}\left(y - \frac{1}{z^2}\right) \\&= xy - \frac{x}{z^2} + \frac{y}{x^2} - \frac{1}{x^2z^2}\end{aligned}$$

Multiply each term in the first bracket by each term in the second. Since all variables are different, no terms can be combined.

Final answer: $xy - \frac{x}{z^2} + \frac{y}{x^2} - \frac{1}{x^2z^2}$

Area-model diagram.

$\times$	y	$-\frac{1}{z^2}$
x	xy	$-\frac{x}{z^2}$
$\frac{1}{x^2}$	$\frac{y}{x^2}$	$-\frac{1}{x^2 z^2}$

Expanding Brackets 25 (a1-025)

Difficulty: Foundation / Marks: 3

Expand and simplify $(a + b)(a^2 - b^2)$.

Worked solution.

Expand the brackets

$$\begin{aligned}
 (a + b)(a^2 - b^2) &= a(a^2 - b^2) + b(a^2 - b^2) \\
 &= a^3 - ab^2 + a^2b - b^3 \\
 &= a^3 + a^2b - ab^2 - b^3
 \end{aligned}$$

Distribute the a and the b across the second bracket. Remember to add the indices for the same bases ($a \times a^2 = a^3$).

Final answer: $a^3 + a^2b - ab^2 - b^3$

Area-model diagram.

$\times$	a^2	$-b^2$
a	a^3	$-ab^2$
b	a^2b	$-b^3$

Expanding Brackets 26 (a1-026)

Difficulty: Foundation / Marks: 4

Expand and simplify $p(p + q) - pq(p^2 - pq)$.

Worked solution.

Expand each part of the expression

$$\begin{aligned}
 p(p + q) - pq(p^2 - pq) &= p(p) + p(q) - pq(p^2) - pq(-pq) \\
 &= p^2 + pq - p^3q + p^2q^2
 \end{aligned}$$

Be careful with the second part; multiplying $-pq$ by $-pq$ results in a positive $+p^2q^2$.

Final answer: $p^2 + pq - p^3q + p^2q^2$

Area-model diagram.

$\times$	p	q
p	p^2	pq

Grid for $p(p + q)$; from this subtract $pq(p^2 - pq) = p^3q - p^2q^2$.

Expanding Brackets 27 (a1-027)

Difficulty: Foundation / Marks: 4

Expand and simplify $\left(\frac{3}{x} - \frac{4}{x^2}\right)(x^2 + 3x)$.

Worked solution.

Distribute the fractional terms

$$\begin{aligned}\left(\frac{3}{x} - \frac{4}{x^2}\right)(x^2 + 3x) &= \frac{3}{x}(x^2 + 3x) - \frac{4}{x^2}(x^2 + 3x) \\ &= 3x + 9 - 4 - \frac{12}{x} \\ &= 3x + 5 - \frac{12}{x}\end{aligned}$$

Multiply each term in the first bracket by each term in the second.

Final answer: $3x + 5 - \frac{12}{x}$

Area-model diagram.

$\times$	x^2	$3x$
$\frac{3}{x}$	$3x$	9
$-\frac{4}{x^2}$	-4	$-\frac{12}{x}$

Expanding Brackets 28 (a1-028)

Difficulty: Foundation / Marks: 3

Expand and simplify $(x^2 - 1)\left(1 + \frac{1}{x}\right)$.

Worked solution.

Expand the brackets

$$\begin{aligned}(x^2 - 1) \left(1 + \frac{1}{x}\right) &= x^2 \left(1 + \frac{1}{x}\right) - 1 \left(1 + \frac{1}{x}\right) \\ &= x^2 + x - 1 - \frac{1}{x}\end{aligned}$$

Multiply through and simplify $x^2 \times \frac{1}{x}$ to get x .

Final answer: $x^2 + x - 1 - \frac{1}{x}$

Area-model diagram.

$\times$	1	$\frac{1}{x}$
x^2	x^2	x
-1	-1	$-\frac{1}{x}$

Expanding Brackets 29 (a1-029)

Difficulty: Foundation / Marks: 4

Expand and simplify $\left(y^2 + \frac{1}{y}\right)(y^3 - 2)$.

Worked solution.

Expand the brackets and simplify the powers

$$\begin{aligned}\left(y^2 + \frac{1}{y}\right)(y^3 - 2) &= y^2(y^3 - 2) + \frac{1}{y}(y^3 - 2) \\ &= y^5 - 2y^2 + \frac{y^3}{y} - \frac{2}{y} \\ &= y^5 - 2y^2 + y^2 - \frac{2}{y} \\ &= y^5 - y^2 - \frac{2}{y}\end{aligned}$$

Multiply both terms in the first bracket by both terms in the second. Simplify $\frac{y^3}{y}$ to y^2 using index laws.

Final answer: $y^5 - y^2 - \frac{2}{y}$

Area-model diagram.

$\times$	y^3	-2
y^2	y^5	$-2y^2$
$\frac{1}{y}$	y^2	$-\frac{2}{y}$

Expanding Brackets 30 (a1-030)

Difficulty: Foundation / Marks: 3

Expand and simplify $2ab(a - b) - a^2(2b - 1)$.

Worked solution.

Expand and simplify

$$\begin{aligned} 2ab(a - b) - a^2(2b - 1) &= 2a^2b - 2ab^2 - 2a^2b + a^2 \\ &= a^2 - 2ab^2 \end{aligned}$$

Distribute each term, then the $2a^2b$ and $-2a^2b$ cancel, leaving $a^2 - 2ab^2$.

Final answer: $a^2 - 2ab^2$

Area-model diagram.

$\times$	a	$-b$
$2ab$	$2a^2b$	$-2ab^2$

Grid for $2ab(a - b)$; from this subtract $a^2(2b - 1) = 2a^2b - a^2$.

Expanding Brackets 31 (a1-031)

Difficulty: Foundation / Marks: 4

Expand and simplify $\left(\frac{x}{2} + 4\right)\left(\frac{4}{x} - 2\right)$.

Worked solution.

Expand and simplify

$$\begin{aligned} \left(\frac{x}{2} + 4\right)\left(\frac{4}{x} - 2\right) &= \frac{x}{2}\left(\frac{4}{x} - 2\right) + 4\left(\frac{4}{x} - 2\right) \\ &= \frac{4x}{2x} - \frac{2x}{2} + \frac{16}{x} - 8 \\ &= 2 - x + \frac{16}{x} - 8 \\ &= \frac{16}{x} - x - 6 \end{aligned}$$

Distribute each term in the first bracket across the second bracket, simplify the fractions, then collect like terms.

Final answer: $\frac{16}{x} - x - 6$

Area-model diagram.

$\times$	$\frac{4}{x}$	-2
$\frac{x}{2}$	2	$-x$
4	$\frac{16}{x}$	-8

Expanding Brackets 32 (a1-032)

Difficulty: Foundation / Marks: 3

Expand and simplify $m^2n(m+n) - n^2m(n+m)$.

Worked solution.

Expand and simplify

$$\begin{aligned} m^2n(m+n) - n^2m(n+m) &= m^3n + m^2n^2 - n^3m - n^2m^2 \\ &= m^3n - mn^3 \end{aligned}$$

Distribute each term, then m^2n^2 and $-n^2m^2$ cancel, leaving $m^3n - mn^3$.

Final answer: $m^3n - mn^3$

Area-model diagram.

$\times$	m	n
m^2n	m^3n	m^2n^2

Grid for $m^2n(m+n)$; from this subtract $n^2m(n+m) = mn^3 + m^2n^2$.

Expanding Brackets 33 (a1-033)

Difficulty: Foundation / Marks: 3

Expand and simplify $\left(1 - \frac{2}{x}\right)(x^2 + 2x)$.

Worked solution.

Expand and simplify

$$\begin{aligned} \left(1 - \frac{2}{x}\right)(x^2 + 2x) &= 1(x^2 + 2x) - \frac{2}{x}(x^2 + 2x) \\ &= x^2 + 2x - 2x - 4 \\ &= x^2 - 4 \end{aligned}$$

Distribute each term. The $2x$ terms cancel, leaving $x^2 - 4$.

Final answer: $x^2 - 4$

Area-model diagram.

$\times$	x^2	$2x$
1	x^2	$2x$
$-\frac{2}{x}$	$-2x$	-4

Expanding Brackets 34 (a1-034)

Difficulty: Foundation / Marks: 3

Expand and simplify $\left(xyz - \frac{y^2}{xz}\right)\left(x^2z^2 + \frac{x}{yz}\right)$.

Worked solution.

Expand the brackets

$$\begin{aligned}
 \left(xyz - \frac{y^2}{xz}\right)\left(x^2z^2 + \frac{x}{yz}\right) &= xyz\left(x^2z^2 + \frac{x}{yz}\right) - \frac{y^2}{xz}\left(x^2z^2 + \frac{x}{yz}\right) \\
 &= x^3yz^3 + \frac{x^2yz}{yz} - \frac{x^2y^2z^2}{xz} - \frac{xy^2}{xyz^2} \\
 &= x^3yz^3 + x^2 - xy^2z - \frac{y}{z^2}
 \end{aligned}$$

Distribute each term. Simplify the second part by dividing each term by x .

Final answer: $x^3yz^3 + x^2 - xy^2z - \frac{y}{z^2}$

Area-model diagram.

$\times$	x^2z^2	$\frac{x}{yz}$
xyz	x^3yz^3	x^2
$-\frac{y^2}{xz}$	$-xy^2z$	$-\frac{y}{z^2}$

Expanding Brackets 35 (a1-035)

Difficulty: Foundation / Marks: 4

Expand and simplify $\left(x - \frac{1}{x}\right) \left(x^2 + 1 + \frac{1}{x^2}\right) \left(x^3 + \frac{1}{x}\right)$.

Worked solution.

Expand the first two brackets, then multiply by the third

$$\begin{aligned} & \left(x - \frac{1}{x}\right) \left(x^2 + 1 + \frac{1}{x^2}\right) \left(x^3 + \frac{1}{x}\right) \\ &= \left[x \left(x^2 + 1 + \frac{1}{x^2}\right) - \frac{1}{x} \left(x^2 + 1 + \frac{1}{x^2}\right) \right] \left(x^3 + \frac{1}{x}\right) \\ &= \left[x^3 + x + \frac{1}{x} - x - \frac{1}{x} - \frac{1}{x^3} \right] \left(x^3 + \frac{1}{x}\right) \\ &= \left(x^3 - \frac{1}{x^3}\right) \left(x^3 + \frac{1}{x}\right) \\ &= x^3 \left(x^3 + \frac{1}{x}\right) - \frac{1}{x^3} \left(x^3 + \frac{1}{x}\right) \\ &= x^6 + x^2 - 1 - \frac{1}{x^4} \end{aligned}$$

Expand the first two brackets — the middle terms cancel giving a difference of cubes. Then multiply the result by the third bracket.

Final answer: $x^6 + x^2 - 1 - \frac{1}{x^4}$

Area-model diagram.

$\times$	x^2	1	$\frac{1}{x^2}$
x	x^3	x	$\frac{1}{x}$
$-\frac{1}{x}$	$-x$	$-\frac{1}{x}$	$-\frac{1}{x^3}$

Grid for the first two brackets; the middle terms cancel to give $x^3 - \frac{1}{x^3}$, which is then multiplied by the third bracket.

Expanding Brackets 36 (a1-036)

Difficulty: Foundation / Marks: 3

Expand and simplify $\left(\frac{2}{y} - 3\right) \left(\frac{y}{2} + 1\right)$.

Worked solution.

Expand the brackets

$$\begin{aligned} \left(\frac{2}{y} - 3\right) \left(\frac{y}{2} + 1\right) &= \frac{2}{y} \left(\frac{y}{2} + 1\right) - 3 \left(\frac{y}{2} + 1\right) \\ &= \frac{2y}{2y} + \frac{2}{y} - \frac{3y}{2} - 3 \\ &= 1 + \frac{2}{y} - \frac{3y}{2} - 3 \\ &= \frac{2}{y} - \frac{3y}{2} - 2 \end{aligned}$$

Multiply the first, outer, inner, and last terms.

Final answer: $\frac{2}{y} - \frac{3y}{2} - 2$

Area-model diagram.

$\times$	$\frac{y}{2}$	1
$\frac{2}{y}$	1	$\frac{2}{y}$
-3	$-\frac{3y}{2}$	-3

Expanding Brackets 37 (a1-037)

Difficulty: Foundation / Marks: 3

Expand and simplify $a^2b(a - b) + ab^2(a + b)$.

Worked solution.

Expand both sets of brackets

$$\begin{aligned}
 a^2b(a - b) + ab^2(a + b) &= (a^3b - a^2b^2) + (a^2b^2 + ab^3) \\
 &= a^3b - a^2b^2 + a^2b^2 + ab^3 \\
 &= a^3b + ab^3
 \end{aligned}$$

Multiply the terms outside by each term inside the brackets.

Final answer: $a^3b + ab^3$

Area-model diagram.

$\times$	a	$-b$
a^2b	a^3b	$-a^2b^2$

Grid for $a^2b(a - b)$; add the analogous grid for $ab^2(a + b) = a^2b^2 + ab^3$.

Expanding Brackets 38 (a1-038)

Difficulty: Foundation / Marks: 2

Expand and simplify $(2x + \frac{3}{x})(2x - \frac{3}{x})$.

Worked solution.

Expand using the difference of two squares identity

$$\begin{aligned}
 \left(2x + \frac{3}{x}\right) \left(2x - \frac{3}{x}\right) &= 2x \left(2x - \frac{3}{x}\right) + \frac{3}{x} \left(2x - \frac{3}{x}\right) \\
 &= 4x^2 - \frac{6x}{x} + \frac{6x}{x} - \frac{9}{x^2} \\
 &= 4x^2 - 6 + 6 - \frac{9}{x^2} \\
 &= 4x^2 - \frac{9}{x^2}
 \end{aligned}$$

Recognize the pattern $(a+b)(a-b) = a^2 - b^2$. Alternatively, expanding fully shows the middle terms cancel out: $4x^2 - \frac{6x}{x} + \frac{6x}{x} - \frac{9}{x^2}$.

Final answer: $4x^2 - \frac{9}{x^2}$

Area-model diagram.

$\times$	$2x$	$-\frac{3}{x}$
$2x$	$4x^2$	-6
$\frac{3}{x}$	6	$-\frac{9}{x^2}$

The off-diagonal terms -6 and $+6$ cancel: a difference of two squares.

Expanding Brackets 39 (a1-039)

Difficulty: Foundation / Marks: 3

Expand and simplify $\left(x + \frac{2}{x}\right)^2$.

Worked solution.

Rewrite as two brackets and expand

$$\begin{aligned}
 \left(x + \frac{2}{x}\right) \left(x + \frac{2}{x}\right) &= x \left(x + \frac{2}{x}\right) + \frac{2}{x} \left(x + \frac{2}{x}\right) \\
 &= x^2 + \frac{2x}{x} + \frac{2x}{x} + \frac{4}{x^2} = x^2 + 2 + 2 + \frac{4}{x^2} = x^2 + 4 + \frac{4}{x^2}
 \end{aligned}$$

Multiply every term in the first bracket by every term in the second bracket.

Final answer: $x^2 + 4 + \frac{4}{x^2}$

Area-model diagram.

$\times$	x	$\frac{2}{x}$
x	x^2	2
$\frac{2}{x}$	2	$\frac{4}{x^2}$

Grid for the perfect square $(x + \frac{2}{x})^2$.

Expanding Brackets 40 (a1-040)

Difficulty: Foundation / Marks: 3

Expand and simplify $(\frac{1}{x^2} - x)(x^3 - x)$.

Worked solution.

Expand the brackets

$$\begin{aligned}
 \left(\frac{1}{x^2} - x\right)(x^3 - x) &= \frac{1}{x^2}(x^3 - x) - x(x^3 - x) \\
 &= \frac{x^3}{x^2} - \frac{x}{x^2} - x^4 + x^2 \\
 &= x - \frac{1}{x} - x^4 + x^2
 \end{aligned}$$

Distribute each term from the first bracket into the second bracket.

Final answer: $-x^4 + x^2 + x - \frac{1}{x}$

Area-model diagram.

$\times$	x^3	$-x$
$\frac{1}{x^2}$	x	$-\frac{1}{x}$
$-x$	$-x^4$	x^2

Expanding Brackets 41 (a1-041)

Difficulty: Foundation / Marks: 4

Expand and simplify $(p - \frac{1}{p})(p + \frac{1}{p})(p^2 + \frac{1}{p^2})$.

Worked solution.

Expand the first two brackets, then multiply by the third

$$\begin{aligned}
 & \left(p - \frac{1}{p}\right) \left(p + \frac{1}{p}\right) \left(p^2 + \frac{1}{p^2}\right) \\
 &= \left[p \left(p + \frac{1}{p}\right) - \frac{1}{p} \left(p + \frac{1}{p}\right)\right] \left(p^2 + \frac{1}{p^2}\right) \\
 &= \left(p^2 - \frac{1}{p^2}\right) \left(p^2 + \frac{1}{p^2}\right) \\
 &= p^2 \left(p^2 + \frac{1}{p^2}\right) - \frac{1}{p^2} \left(p^2 + \frac{1}{p^2}\right) \\
 &= p^4 + 1 - 1 - \frac{1}{p^4} \\
 &= p^4 - \frac{1}{p^4}
 \end{aligned}$$

The first two brackets form a difference of two squares. The result multiplied by the third bracket forms another difference of two squares.

Final answer: $p^4 - \frac{1}{p^4}$

Area-model diagram.

$\times$	p	$\frac{1}{p}$
p	p^2	1
$-\frac{1}{p}$	-1	$-\frac{1}{p^2}$

Grid for the first two brackets giving $p^2 - \frac{1}{p^2}$; multiplying by $(p^2 + \frac{1}{p^2})$ repeats the difference-of-squares pattern.

Expanding Brackets 42 (a1-042)

Difficulty: Foundation / Marks: 4

Expand and simplify $p(p + q) - (p - q)(p + q)$.

Worked solution.

Expand the first single bracket, and then expand the pair of brackets.

$$\begin{aligned}
 p(p + q) - (p - q)(p + q) &= p^2 + pq - p(p + q) + q(p + q) \\
 &= p^2 + pq - p^2 - pq + pq + q^2 \\
 &= q^2 + pq
 \end{aligned}$$

Recognize the difference of two squares. The middle terms cancel out.

Final answer: $q^2 + pq$

Area-model diagram.

$\times$	p	q
p	p^2	pq
$-q$	$-pq$	$-q^2$

Grid for $(p - q)(p + q) = p^2 - q^2$; this is subtracted from $p(p + q) = p^2 + pq$.

Expanding Brackets 43 (a1-043)

Difficulty: Foundation / Marks: 4

Expand and simplify $(x + y)^3$.

Worked solution.

Expand the first two brackets, and then expand the final bracket.

$$\begin{aligned}
 (x + y)^3 &= (x + y)(x + y)(x + y) = \left((x + y)(x + y)\right)(x + y) \\
 &= \left(x(x + y) + y(x + y)\right)(x + y) \\
 &= \left(x^2 + xy + xy + y^2\right)(x + y) \\
 &= \left(x^2 + 2xy + y^2\right)(x + y) \\
 &= x(x^2 + 2xy + y^2) + y(x^2 + 2xy + y^2) \\
 &= x^3 + 2x^2y + xy^2 + x^2y + 2xy^2 + y^3 \\
 &= x^3 + 3x^2y + 3xy^2 + y^3
 \end{aligned}$$

Recognize the difference of two squares. The middle terms cancel out.

Final answer: $x^3 + 3x^2y + 3xy^2 + y^3$

Area-model diagram.

$\times$	x	y
x	x^2	xy
y	xy	y^2

Grid for $(x + y)^2$; the result $x^2 + 2xy + y^2$ is then multiplied by $(x + y)$.

Expanding Brackets 44 (a1-044)

Difficulty: Foundation / Marks: 4

Expand and simplify $(ab + cd)(ac - bd)$.

Worked solution.

Expand the brackets

$$\begin{aligned}(ab + cd)(ac - bd) &= ab(ac - bd) + cd(ac - bd) \\ &= a^2bc - ab^2d + ac^2d - bcd^2\end{aligned}$$

Multiply each term in the first bracket by each term in the second bracket.

Final answer: $a^2bc - ab^2d + ac^2d - bcd^2$

Area-model diagram.

$\times$	ac	$-bd$
ab	a^2bc	$-ab^2d$
cd	ac^2d	$-bcd^2$

Expanding Brackets 45 (a1-045)

Difficulty: Foundation / Marks: 4

Expand and simplify $(2x - 1)^2(x + 3)$.

Worked solution.

Expand the brackets

$$\begin{aligned}(2x - 1)^2(x + 3) &= (2x - 1)(2x - 1)(x + 3) \\ &= (2x(2x - 1) - 1(2x - 1))(x + 3) \\ &= (4x^2 - 2x - 2x + 1)(x + 3) \\ &= (4x^2 - 4x + 1)(x + 3) \\ &= 4x^2(x + 3) - 4x(x + 3) + 1(x + 3) \\ &= 4x^3 + 12x^2 - 4x^2 - 12x + x + 3 \\ &= 4x^3 + 8x^2 - 11x + 3\end{aligned}$$

Multiply each term in the first bracket by each term in the second bracket.

Final answer: $4x^3 + 8x^2 - 11x + 3$

Area-model diagram.

$\times$	$2x$	-1
$2x$	$4x^2$	$-2x$
-1	$-2x$	1

Grid for $(2x - 1)^2$; the result $4x^2 - 4x + 1$ is then multiplied by $(x + 3)$.
